

AMMS2603 DISCRETE MATHEMATICS

TEST

Answer all the questions.

- Q1. Let $U = \{x \mid x \text{ is an integer and } 0 < 5x \leq 50\}$ be the universal set that contains the following subsets:

$1, 2, 3, 4, 5, 6, 7, 8, 9, 10$

$A = \{x \mid x \text{ is a prime number}\},$

$B = \{x \mid x \text{ is a factor of } 18\}$ and

$C = \{x \mid x \text{ is a multiple of } 2\}.$

- (a) List all elements of U, A, B and C . (4 marks)
- (b) Find $(A \cap B) \cup C$ and $\bar{A} \oplus C$. (6 marks)
- Q2. Find the greatest common divisor (GCD) and the least common multiple (LCM) of 5245 and 930 by using the Euclidean algorithm. (6 marks)
- Q3. Let set $A = \{1, 2, 3, 4\}$. A relation R on A is defined by xRy if and only if $x > y-1$.
- (a) List the ordered pairs which belong to the relation R . (5 marks)
- (b) Find the domain and range of R . (2 marks)
- (c) Construct the matrix of R . (3 marks)
- (d) Draw the digraph for R and list the in-degrees and out-degrees of each vertex. (4 marks)
- Q4. Let $A = p \leftrightarrow [(p \wedge \sim q) \rightarrow r]$.
- (a) Construct a truth table for the expression A . Hence determine whether the expression A is a tautology, contradiction, or contingency. (5 marks)
- (b) Obtain the Principal Disjunctive Normal Form (PDNF) and Principal Conjunctive Normal Form (PCNF) of A . Hence, deduce the PDNF and PCNF of $\sim A$. (6 marks)

Q5. Find the negation of the following statement:

$$(p \rightarrow \sim q) \vee p$$

and simplify your answer to the simplest form.

(5 marks)

Q6. State the scope of each quantifier in the following expression:

$$\text{"}\exists x, [\sim Q(x, y) \wedge \forall y (R(x, y) \wedge P(x, y))\text{"}$$

and state whether the expression above is a statement. Explain your answer. (4 marks)

[Total: 50 marks]

Logical Equivalences:

- | | |
|---|-------------------|
| 1) $P \wedge T \equiv P$
$P \vee F \equiv P$ | Identity laws |
| 2) $P \vee T \equiv T$
$P \wedge F \equiv F$ | Domination laws |
| 3) $P \vee P \equiv P$
$P \wedge P \equiv P$ | Idempotent laws |
| 4) $\sim(\sim P) \equiv P$ | Double negation |
| 5) $P \vee Q \equiv Q \vee P$
$P \wedge Q \equiv Q \wedge P$ | Commutative laws |
| 6) $P \vee (Q \vee R) \equiv (P \vee Q) \vee R$
$P \wedge (Q \wedge R) \equiv (P \wedge Q) \wedge R$ | Associative laws |
| 7) $P \vee (Q \wedge R) \equiv (P \vee Q) \wedge (P \vee R)$
$P \wedge (Q \vee R) \equiv (P \wedge Q) \vee (P \wedge R)$ | Distributive laws |
| 8) $\sim(P \wedge Q) \equiv \sim P \vee \sim Q$
$\sim(P \vee Q) \equiv \sim P \wedge \sim Q$ | De Morgan's law |
| 9) $P \rightarrow Q \equiv \sim P \vee Q$
$P \wedge \sim P \equiv F$
$P \vee \sim P \equiv T$ | |